



Power Performance Histories of Radioisotope Thermoelectric Generators, Their Robust Mission Service, and Observations About Long-Lived, RTG-Powered Missions

David Woerner

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California

Christopher Whiting, Allen Guzik, Lauren Clayman, and Leah Sopko

Glenn Research Center, Cleveland, Ohio

June Zakrajsek

The Aerospace Corporation, El Segundo, California

NASA STI Program Report Series

Since its founding, NASA has been dedicated to the advancement of aeronautics and space science. The NASA scientific and technical information (STI) program plays a key part in helping NASA maintain this important role.

The NASA STI program operates under the auspices of the Agency Chief Information Officer. It collects, organizes, provides for archiving, and disseminates NASA's STI. The NASA STI program provides access to the NTRS Registered and its public interface, the NASA Technical Reports Server, thus providing one of the largest collections of aeronautical and space science STI in the world. Results are published in both non-NASA channels and by NASA in the NASA STI Report Series, which includes the following report types:

- **TECHNICAL PUBLICATION.**
Reports of completed research or a major significant phase of research that present the results of NASA programs and include extensive data or theoretical analysis. Includes compilations of significant scientific and technical data and information deemed to be of continuing reference value. NASA counterpart of peer-reviewed formal professional papers but has less stringent limitations on manuscript length and extent of graphic presentations.
- **TECHNICAL MEMORANDUM.**
Scientific and technical findings that are preliminary or of specialized interest, e.g., quick release reports, working papers, and bibliographies that contain

minimal annotation. Does not contain extensive analysis.

- **CONTRACTOR REPORT.**
Scientific and technical findings by NASA-sponsored contractors and grantees.
- **CONFERENCE PUBLICATION.**
Collected papers from scientific and technical conferences, symposia, seminars, or other meetings sponsored or cosponsored by NASA.
- **SPECIAL PUBLICATION.**
Scientific, technical, or historical information from NASA programs, projects, and missions, often concerned with subjects having substantial public interest.
- **TECHNICAL TRANSLATION.**
English-language translations of foreign scientific and technical material pertinent to NASA's mission.

Specialized services also include organizing and publishing research results, distributing specialized research announcements and feeds, providing information desk and personal search support, and enabling data exchange services.

For more information about the NASA STI program, see the following:

- Access the NASA STI program home page at <http://www.sti.nasa.gov>



Power Performance Histories of Radioisotope Thermoelectric Generators, Their Robust Mission Service, and Observations About Long-Lived, RTG-Powered Missions

David Woerner

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California

Christofer Whiting, Allen Guzik, Lauren Clayman, and Leah Sopko

Glenn Research Center, Cleveland, Ohio

June Zakrajsek

The Aerospace Corporation, El Segundo, California

National Aeronautics and
Space Administration

Glenn Research Center
Cleveland, Ohio 44135

Acknowledgments

The authors of this report thank Phil Ugorowski for his review of this report and his comments, and Tom Hammel and Russel Bennett for their review and advice regarding behavior of the PbTe based thermoelectric materials. The authors are particularly grateful to Dr. Ralph McNutt, who was instrumental in collecting and consolidating a significant amount of the oldest RTG performance data. The authors built upon that beginning to ultimately provide the performance dataset presented here.

A portion of this research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004).

Level of Review: This material has been technically reviewed by technical management.

This report is available in electronic form at <https://www.sti.nasa.gov/> and <https://ntrs.nasa.gov/>

NASA STI Program/Mail Stop 050
NASA Langley Research Center
Hampton, VA 23681-2199

Power Performance Histories of Radioisotope Thermoelectric Generators, Their Robust Mission Service, and Observations About Long-Lived, RTG-Powered Missions

David Woerner
National Aeronautics and Space Administration
Jet Propulsion Laboratory
California Institute of Technology
Pasadena, California 91109

Christofer Whiting, Allen Guzik, Lauren Clayman, and Leah Sopko
National Aeronautics and Space Administration
Glenn Research Center
Cleveland, Ohio 44135

June Zakrajsek
The Aerospace Corporation
El Segundo, California 90245

Motivation

Proposers from spaceflight centers would like to submit mission concepts powered by Radioisotope Thermoelectric Generators (RTGs) with primary mission durations longer than 14 years to NASA. This white paper begins to address these requests from mission proposers and provides a summary of historical data on missions that use(d) RTGs. The data shows all RTGs flown by the United States have performed past their design life and that a majority have flown well past their design life. Many of these thermoelectric systems have lasted longer than the current 14-year flight design life requirement of the available Multi-Mission RTG (MMRTG).

Background

Radioisotope Thermoelectric Generators (RTGs) have enabled pioneering missions for scientific exploration of deep space, Mars, and the Moon. The United States has improved and flown RTGs for the last 60+ years. Most of these systems are based on technological advances completed in the 1960s, 70s, and 80s. These advances have proven to be extremely reliable, outlasting their original design life, as in the case of the Voyager missions, which are still utilizing six five-decades-old Multi-Hundred Watt RTGs launched in 1977. As of this publication, all flown U.S. RTGs have operated beyond their primary mission duration and none has caused a mission failure.

It is essential to make a clear distinction between an RTG's design life and mission duration. Engineers design RTGs to meet a performance benchmark known as the design life. Despite being referred to as "design life," this design parameter does not demarcate certain failure, nor does it signify an increase in risk of RTG failure. As of this writing, NASA has imposed Radioisotope Power Systems (RPS) design life requirements as up to 3 years fueled and in storage plus 14 years of design life in flight

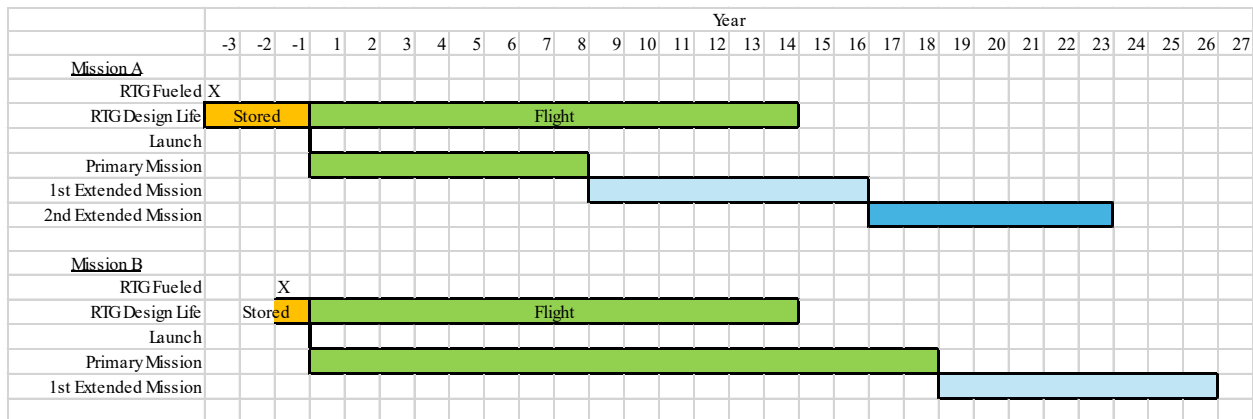


Figure 1.—Hypothetical mission timelines. Mission A concludes its primary mission within the flight segment or an RTG's design life. Mission B's primary mission extends beyond the length of the flight segment of an RTG's design life.

for a total design life of 17 years. See Figure 1. This paper is taking these design life requirements as is and does not describe the basis for, nor challenge, those requirements. An RTG's design life begins when it is fueled. The end of the 14-year in-flight segment is called the End of Design Life (EODL). An RTG's design life is about establishing requirements for the engineering analyses completed during an RTG's development. EODL is not a point at which an RTG will fail; it is the duration used in design analyses. RPS engineers, as electrical circuit designers, build in conservatism so the design life requirement will be met and exceeded.

Design life is not the same as mission duration. Mission designers and planners are at liberty to choose primary mission durations that allow the exploration they are advocating for to be completed completely independently of the RPS's design life. To date, however, mission designers have felt constrained to not allow a primary mission's duration to exceed an RTG's design life. This assumption inappropriately conflates the EODL engineering benchmark with the actual life of the RTG.

NASA cannot guarantee specific performance metrics beyond the design requirements of 14 years in flight for the currently available generators. It can, however, publish historical data about the in-flight performance of these systems and discuss the risk they may fail catastrophically, or fail to perform adequately, after operating past their design life. That is the aim of this white paper.

Another important nuance is primary mission duration contrasted with extended mission duration. A primary mission duration is the length of time a team of scientists and spacecraft designers have negotiated with their funding organization to complete the science and mission objectives and requirements. A first extended mission, if funded and approved, begins when the primary mission ends. Multiple extensions to missions are possible. Each extended mission may have a unique set of operational requirements. For the purposes of comparing mission durations with RTG design life requirements in this paper, design life requirements apply to primary mission duration. Total mission duration includes primary and extended mission durations. See column, "Mission Duration," to see this distinction in Table 1 and Table 2. Figure 1 depicts hypothetical mission timelines alongside RTG design life. It is not representative of any actual or planned missions. Extended missions are not a given for any NASA mission.

TABLE 1.—MISSIONS USING SiGe-BASED RTGs. THE UNITED STATES HAS FLOWN FIFTEEN SiGe RTGs ON SPACECRAFT WHOSE SUCCESSFUL MISSION DURATIONS EXCEEDED 15 YEARS. THIRTEEN OF 17 SiGe RTGs HAVE FLOWN FOR 20+ YEARS.

Mission	Launch Date	RTG Details			Mission Duration ^{a,b}		Mission Elapsed Time of Ongoing Missions as of: 10/1/2024
		Type	Number	Requirements (per RTG)	Primary	Total	
LES 8	March 15, 1976	MHW-RTG	2	125 W at 5 years	5 years	28 years 3 months	N/A
LES 9	March 15, 1976	MHW-RTG	2	125 W at 5 years	5 years	44 years 2 months	N/A
Voyager 2	August 20, 1977	MHW-RTG	3	128 W at 4 years	4 years	Ongoing	47 years, 1 month
Voyager 1	September 5, 1977	MHW-RTG	3	128 W at 4 years	4 years	Ongoing	47 years, 1 month
Galileo	October 18, 1989	GPHS-RTG	2	235 W at 8.1 years	8 years	13 years 11 months	N/A
Ulysses	October 6, 1990	GPHS-RTG	1	245 W at 4.8 years	5 years	18 years 9 months	N/A
Cassini	October 15, 1997	GPHS-RTG	3	199 W at 16 years	11 years	19 years 11 months	N/A
New Horizons	January 19, 2006	GPHS-RTG	1	191 W at 9.5 years	10 years	Ongoing	18 years, 9 months

^aIncludes time to travel to the mission objective for non-surface missions (i.e., Galileo's mission began at launch whereas Viking's mission began upon landing on Mars).

^bIt can take months to download data from a spacecraft, so the official end of a primary mission is not well defined. Most primary mission durations are rounded to the nearest year for simplicity.

TABLE 2.—MISSIONS USING PbTe-BASED RTGs. NASA HAS FLOWN NINE PbTe RTGs ON SPACECRAFT WHOSE SUCCESSFUL MISSION DURATIONS EXCEEDED 15 YEARS. EIGHT OF 30 PbTe RTGs HAVE FLOWN FOR 20+ YEARS.

Mission	Launch Date	RTG Details			Mission Duration ^{a,b}		Mission Elapsed Time of Ongoing Missions as of: 10/1/2024
		Type	Number	Requirements (per RTG)	Primary	Total	
Transit 4-A	June 29, 1961	SNAP-3B	1	2.7 W at BOL with 5-year design life	N/A ^c	17+ years ^d	N/A
Transit 4-B	November 15, 1961	SNAP-3B	1	2.7 W at BOL with 5-year design life	N/A ^c	9 months	N/A
Transit 5BN-1	September 28, 1963	SNAP-9A	1	25 W at 5 years	N/A ^c	8 months	N/A
Transit 5BN-2	December 5, 1963	SNAP-9A	1	25 W at 5 years	N/A ^c	1 year 7 months	N/A
Transit 5BN-3	April 21, 1964	SNAP-9A	1	25 W at 5 years	N/A ^c	Launch Failure	N/A
Nimbus B	May 18, 1968	SNAP-19B	2	25 W at 1 year	1 year	Launch Failure	N/A
Nimbus III	April 14, 1969	SNAP-19B	2	25 W at 1 year	1 year	2 years 9 months	N/A
Apollo 12	November 14, 1969	SNAP-27	1	63.5 W at 1 year	1 year	7 years 11 months	N/A
Apollo 13	April 11, 1970	SNAP-27	1	63.5 W at 1 year	1 year	Mission Abort	N/A
Apollo 14	January 31, 1971	SNAP-27	1	63.5 W at 1 year	1 year	6 years 10 months	N/A
Apollo 15	July 26, 1971	SNAP-27	1	63.5 W at 1 year	1 year	6 years 1 months	N/A
Pioneer 10	March 2, 1972	SNAP-19	4	30 W at 3 years	3 years	30 years 11 months	N/A
Apollo 16	April 16, 1972	SNAP-27	1	63.5 W at 1 year	1 year	5 years 2 months	N/A
TRIAD	September 2, 1972	SNAP-19	1	30 W at 5 years	N/A ^c	10+ years ^d	N/A
Apollo 17	December 7, 1972	SNAP-27	1	69 W at 2 years	2 years	4 years 5 months	N/A
Pioneer 11	April 5, 1973	SNAP-19	4	30 W at 3 years	3 years	22 years 8 months	N/A
Viking 1	August 20, 1975	SNAP-19	2	35 W at 90 days after landing on Mars	90 days	6 years 4 months	N/A
Viking 2	September 9, 1975	SNAP-19	2	35 W at 90 days after landing on Mars	90 days	3 years 11 months	N/A
Curiosity	November 26, 2011	MMRTG	1	14-year flight design life	2 years	Ongoing	12 years, 10 months
Perseverance	July 30, 2020	MMRTG	1	14-year flight design life	2 years	Ongoing	4 years, 2 months

^aIncludes time to travel to the mission objective for non-surface missions (i.e., Galileo's mission began at launch whereas Viking's mission began upon landing on Mars).

^bIt can take months to download data from a spacecraft, so the official end of a primary mission is not well defined. Most primary mission durations are rounded to the nearest year for simplicity.

^cExpected primary mission durations are not discussed for these DoD missions.

^dThe exact end date for this mission is not publicly available. The duration represents the most recent known confirmation of the mission still operating.

Supporting Evidence and Analysis

No correlation has been found between RTG design life and RTG failures. In fact, no RTG has failed in flight before mission failure, termination, or decommissioning. The United States has chosen design enhancements that increase an RTG's lifetime based upon observations from RTGs that were developed, tested, and flown in the last 60+ years, along with design trade studies and laboratory experiments. Significant time and money has been invested by the U.S. Government focused on designing the systems to provide reliable power to enable unique and valuable space exploration and scientific discoveries. The historical record reflects this.

It is important to note that NASA has flown two different classes of RTG designs, distinguished by the type of thermoelectric materials used and their mechanical features. RTGs developed at the dawn of the space race, as well as the current MMRTG, used thermoelectric couples based on a PbTe (lead telluride) thermoelectric system of materials. The Multi-Hundred Watt-RTG (MHW-RTG) and General-Purpose Heat Source-RTG (GPHS-RTG) relied on conversion materials composed of SiGe (silicon germanium). NASA's new Next Generation Mod 1 RTG also uses the SiGe thermoelectric couple, also known as a unicouple. Both generators, using these two material systems, are long-lived but configured and operated differently.

PbTe thermoelectric couple systems rely on spring-loading to hold the PbTe thermoelectric couples in place while the SiGe thermoelectric couple systems fasten each unicouple in place as a cantilever. The PbTe systems operate with hot-junction temperatures near 500 °C, while the SiGe systems operate near 1,000 °C. Lists of potential failure modes for these two classes of RTG systems have some overlap (e.g., too many broken couples would cause the electrical circuit to fail open), but each has some unique potential failure modes imposed by the chemical differences of thermoelectric couples and the design of the RTGs. For example, a micrometeorite puncturing an RTG housing enclosing PbTe thermoelectric couples will cause the loss of cover gases suppressing thermoelectric material sublimation. Unsuppressed sublimation will degrade the performance of this type of RTG and cause the RTG to fail in a short period. However, SiGe-based RTGs operate in a vacuum and do not need cover gases, thus avoiding this risk.

U.S. agencies have shared lessons learned and techniques to improve the reliability of RTGs. The in-flight record of RTGs has revealed no distinction in the lifetime between PbTe and SiGe-based RTGs since the Pioneer 10 and Voyager 2 launches. Despite shorter average flight/mission durations compared to SiGe-based RTGs, PbTe-based RTGs have served on some missions lasting over 30 years. While both systems are long-lived, the performance and degradation of these two types of RTGs do not compare directly, as seen in Table 1 and Table 2. The requirements on the flight portion of design life for each RTG are listed in the columns labeled "Requirements (per RTG)" in Table 1 and Table 2.

It is important to note that the spacecraft listed in Table 1 and Table 2 have mission durations measured from their launch dates to the time of last communication. In part, humankind has lost the actual power data from some of those missions. For example, Pioneer 10 operated for over 30 years and NASA could communicate with it, but NASA did not record the data from the RTGs for the last 6 years because funding had ended to monitor the spacecraft. Eventually, communications became too weak to be detected. LES 8 and 9 operated for 28 and 44 years, respectively. The RTG data available to the public ends in 1996, yet the spacecraft communicated with operators until decommissioning.

The past performance of RTGs in the plot in Appendix A shows and hides many important details. Almost all the missions (especially the earliest ones) noted in this plot had short primary mission durations. This gave each mission some margin, margin the mission designers may have been unaware of because America was still gaining confidence in and maximizing the design life of the generators. For example, when the two Voyager spacecraft launched, each had a planned primary mission of 4 years. The RTGs they carried had the same design life. Now, with 47+ years in space, it is obvious the RTGs had a significant design life margin. In addition, Pioneer 10 and 11 had primary mission durations of 36 months and a probabilistic requirement of 0.995 on each RTG to produce 30 W at Jupiter encounter (with four SNAP-19 RTGs per spacecraft). NASA timed the Jovian encounter to be 36 months after delivery of the RTGs to NASA. One of the two Pioneer spacecraft ultimately flew over 30 years in space. The other flew over 22 years.

NASA flew most of the missions listed in Table 1 and Table 2 in benign environments. The value of this point is paramount. The Voyager, Pioneer, and New Horizons missions flew away from Earth on trajectories where the spacecraft operated in stable thermal environments, avoiding the potential life-limiting effect of thermal cycles. Galileo and Cassini traveled away from the sun after performing a Venus flyby, and experienced small thermal cycles in orbit about their planetary targets. The mission designers selected trajectories, destinations, and other variables with environments that did not impose large thermal cycles induced by their mission trajectories, or when they did, they were brief (e.g., Venus flybys). In addition, the mission designers did not subject the RTGs to frequent and high amplitude vibrations after the launch. Behind this success was a backdrop of thermal, radiation, and other environmental testing of parts in the RTGs to support the design life analyses and choices mission designers made. Based on these numerous past successes, mission design can be used to qualitatively and quantitatively manage risks associated with durations that exceed an RTG's design life.

We must note that the capability to manufacture SiGe uncouples for the soon-to-be built Next Generation Mod 1 RTGs is nascent. That production capability is being engineered using the drawings, procedures, and processes used when GPHS-RTGs were still being made. The test and manufacturing results from that period are being used to guide the restart of SiGe uncouple production. Although not proven, the new uncouples, which are based on the same historical design, are expected to have the same performance, including degradation and life. However, long-duration testing will be required to prove this assumption.

Further, NASA missions and programs have a long history of a close partnership with the Department of Energy and its industrial contractors in developing and qualifying RTG technologies. This is especially true in Independent Verification and Validation (IV&V), which NASA/JPL has been supporting since at least 1965 during both the technology and flight system development phases. That support has relied on a unique set of capabilities and expertise, which includes long-term performance testing of materials, components, and electrically heated thermoelectric generators (ETGs). Results from those and other tests were and are used to anchor physics-based, system-level performance models capable of predicting RTG power output across a range of mission scenarios, environments, and operating conditions and are now typically used during mission design and in flight. This support was initiated when the Pioneer and Viking missions were being readied for flight and has continued to today with MSL, M2020, and additional missions in development.

The reader should, of course, recognize and assume that time is a life-limiting phenomenon. To date, however, no catastrophic RTG failure has been seen in flight. NASA understands the historical data and degradation of RTG power from PbTe and SiGe RTGs sufficiently to project power output for very long mission durations. To be clear, a quantified risk has not been established for RTG failures for RTGs operating beyond their required design life. Beginning with the Pioneer 10 and Voyager 2 launches, the historical data from RTG-powered missions shows slow performance degradation, see Appendix A.

When planning an RTG-powered mission with a duration of greater than 14 years, and when the mission environments emulate previous long-lived RTG missions, there is little or no measurable risk incurred. In fact, RTG power data does not signal or indicate a risk for designing mission flight durations of 20 years or more. Evidence in the flight data of modern RTGs shows power output to be predictable and reliable when operated well past an RTG's design life. Appendix B cites some examples of proposed missions that would fly longer than 14 years.

Conclusion

Mission designers have felt constrained to not allow a primary mission's duration to exceed a Radioisotope Thermoelectric Generator's (RTG's) design life. The Radioisotope Power Systems (RPS) Program has evaluated the historical record, as summarized in this paper, and qualitatively observes that the record supports mission durations that exceed 14 years in flight when rigorously considering environmental effects in a mission design. Every RTG on a U.S. mission has operated beyond their primary mission duration. RTGs do not abruptly stop supplying power when they reach their End of Design Life (EODL) and, in general, they degrade gracefully.

Appendix A.—U.S. RTG Normalized Performance

Figure 2 plots the actual normalized performance data from most of the RTGs flown by the United States. It plots the ratio of actual power to the power measured when a mission begins (i.e., power output shortly after launch, when an RTG is thermally stable). This form allows for an easy comparison of the curves. This plot uses actual flight data and does not make assumptions or include any extrapolations.

Figure 2 is more illustrative than informative. No format can present all this data in a way that allows nuances in performance to be seen. Some missions only lasted months (Transit 4), while others are still active after 47 years (Voyagers). Some missions show a gradual degradation of power, resulting in a curve that declines gently for decades. Other missions show rapid RTG degradation. Figure 3 shows this distinction by separating the missions based on their degradation behavior. A discussion of this rapid degradation follows Figure 3.

The power data in Figure 2 for a few missions are incomplete. This includes Transit 5BN-2 after its primary function as a navigational satellite ended, Pioneer 10 after 12/31/95, and LES 8 and 9 after 9/1/88.

Performance data for the Transit 4-A, Transit 4-B, and TRIAD missions is not included. The Transit 4 satellites did not telemeter RTG power output, and it is unclear if this was the limit of the RTG or the spacecraft. The small, 2.7 We power sources, or SNAP-3Bs, were the first RTGs to fly in space and provided supplementary power to the Transit 4-A and Transit 4-B satellites.

A single Transit-RTG powered the TRIAD navigational satellite, which was launched in September 1972. Unfortunately, a failure in the satellite prevented transmitting any RTG data after 30 days. Some experiments onboard the satellite were still usable, however, so the mission continued. The only publicly available performance data on the Transit-RTG are average power values over the first 20 days and average power from 25 to 30 days.

A final note, NASA does not have SiGe-based RTG degradation data beyond the current flight durations of the Voyager RTGs, although the physics-based models can project performance beyond 47 years. Similarly, PbTe-based RTG degradation data does not exist past the recorded Pioneer 10 data (~24 years).

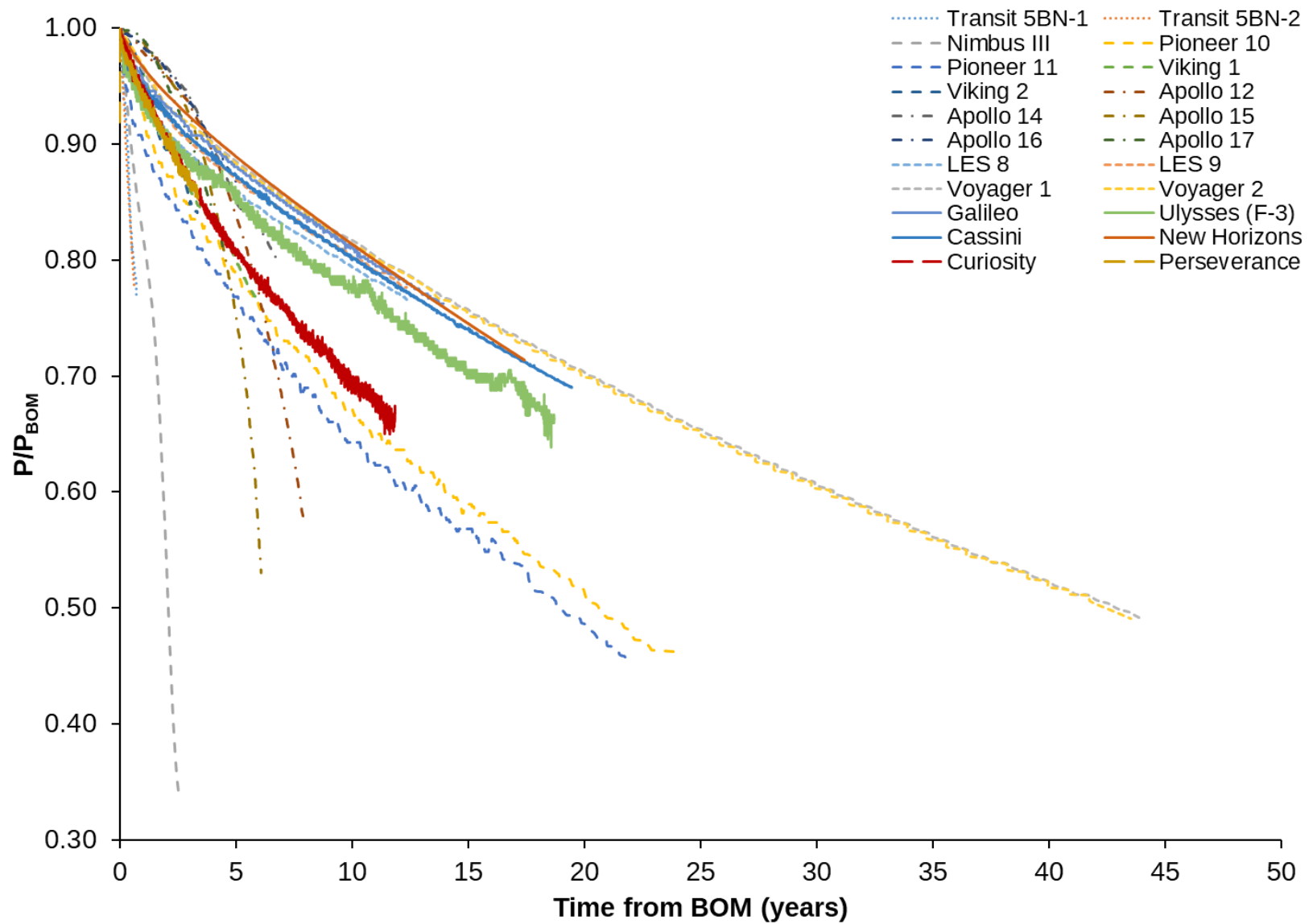


Figure 2.—U.S. RTG Normalized Performance Plot.

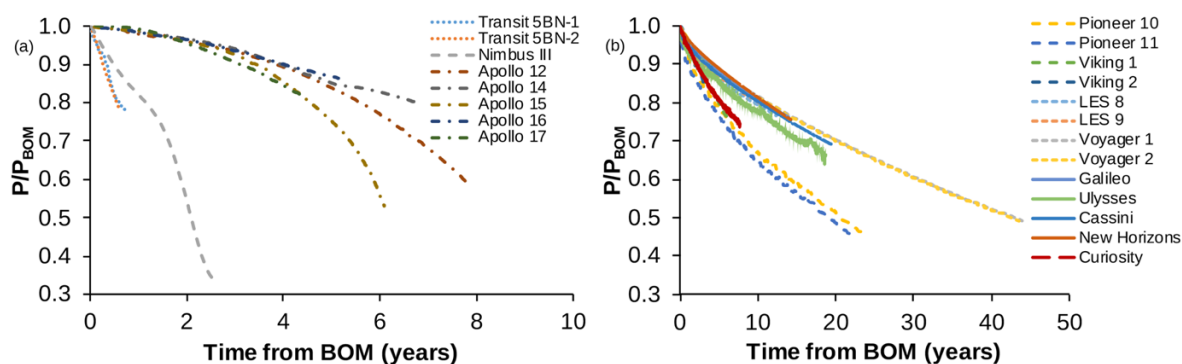


Figure 3.—(a) Life performance for RTG missions that predate the first flight of the SNAP-19 RTG (the SNAP-19B predated the flight of the SNAP-19). (b) Life performance for RTG missions that flew with, or after, the SNAP-19 RTG's development.

Figure 3(a) displays the performance data of early RTGs that predate the SNAP-19, revealing the high degradation rates observed in those early systems. Lessons learned from each of those designs resulted in improvements in subsequent designs that decreased degradation and increased their operating duration for missions, Figure 3(b).

Specifically, both the SNAP-9A (Transit 5BN missions) and SNAP-19B (Nimbus III) observed issues with their gas management system. PbTe-based thermoelectric systems require a cover gas to suppress sublimation. Available literature does not discuss this issue in more detail for the SNAP-9A. It is therefore suspected that the 36 O-rings used to seal the thermoelectric modules into the generator could not maintain a sufficient cover gas to suppress sublimation. Nimbus III experienced a similar problem even though the design only possessed O-ring seals at the generator end-caps. Analysis suggested that the 15 psia argon cover gas would slowly leak past the O-rings over several years. Helium, generated by the ^{238}Pu fuel, would replenish the cover gas, achieving an equilibrium pressure of 2 to 5 psia. The analysis proved to be incorrect. Pressure sensors in the SNAP-19B measured only 0.2 psia after 9 months, and after 1.1 years the cover gas was no longer sufficient to suppress rapid sublimation. These gas management issues led to a change in design philosophy. All subsequent PbTe-based RTGs house the thermoelectric couples in a hermetically sealed cavity. Since then, no gas management issues have appeared.

The SNAP-27 RTGs (Apollo missions) also degraded faster than currently desired. This was because of conscious design choices and a higher than predicted lunar surface temperature. Engineers designed the SNAP-27 RTG for a 1-year mission. In an effort to obtain more power, new PbTe thermoelectric materials were used at higher operating temperatures. Engineers decided to operate SNAP-27 near its maximum operating temperature to maximize power for this short mission. Design analysis indicated a hot junction temperature of 866 K during the lunar day. Results from the Apollo 12 mission, however, reported SNAP-27 hot junction temperatures that ranged from 890 to 855 K. Performance degradation of the deployed SNAP-27s was slower than, but similar to, accelerated aging tests performed with hot side temperatures of 933 and 1005 K. Modern PbTe thermoelectric experts have reviewed the performance of the SNAP-27 RTGs and concluded that the behavior suggests thermal runaway. Thermal runaway is a phenomenon when sublimation causes the cross-section of the thermoelectric couples to shrink, leading to increased hot side temperatures. If these temperature increases are faster than the decrease in heat because of decay of the fuel, then the sublimation rate will increase, which causes a more rapid increase in temperature.

Two perspectives illuminate the SNAP-27's performance. On one hand, the SNAP-27 RTG shows what can happen when designers use new materials without fully understanding their long-term performance and design systems with little or no operational margin (temperature, in this case). On the other hand, it is a testament to high-quality engineering as the RTGs experienced 1 to 2% power loss over the original 1-year mission requirement, which outperforms any other generators flown to date (modern generators lose ~5% over the first year and quickly level out afterwards).

Today, RTG design philosophy focuses on relatively long-life and robustness for NASA missions. Lessons learned from SNAP-9A, SNAP-19B, and SNAP-27 were key to assuring that modern and future generators will provide the reliable performance that the RTG user community has come to expect.

Reference

C.E. Whiting and D.F. Woerner, "Chapter 8: Lifetime Performance of Spaceborne RTGs," The Technology of Discovery: Radioisotope Thermoelectric Generators and Thermoelectric Technologies for Space Exploration, D.F. Woerner, Ed., John Wiley & Sons, 2023. ISBN: 9781119811367.
Reference data source: <https://datadryad.org/stash/dataset/doi:10.5061/dryad.1zcrjdfw2>

Appendix B.—Examples of Proposals Needing Power Sources With a Design Life Over 14 Years

Table B.1 gives examples of proposals needing power sources with a design life greater than 14 years in flight. This list of examples is not comprehensive, but illustrative. The table calculates Mission Duration as the sum of Cruise Duration and Science Capture Duration.

TABLE B.1.—PROPOSALS NEEDING POWER SOURCES WITH DESIGN LIFE OVER 14 YEARS

Mission	Organization	Study Date (Mon-yr)	Class	Target	Mission Duration (yr)	Cruise Duration (yr)	Science Duration (yr)	Power Level BOM (W _e)	Power Level EOM (W _e)
Uranus Orbiter (CAELUS)	JPL	Aug-14	New Frontiers	Uranus	14.8	14	0.8	334	231
Uranus Orbiter and Probe - NPAS - RTG	APL	Sep-14	Flagship	Uranus	15.4	13	2.4	496	379
Uranus Orbiter and Probe - NPAS - SRG	APL	Sep-14	Flagship	Uranus	15.4	13	2.4	720	588
Neptune Vision	JPL	2005	Flagship	Neptune	15	12	3	916	688
Enceladus Sample Return (4)	JPL	Apr-10	Flagship	Enceladus	16	16	0.01	480	-----
Neptune-Triton High Power Orbiter (1)	JPL	Feb-10	Flagship	Neptune	15	13	2	700	-----
Ice Giant Study: Neptune Orbiter + Probe	JPL	Jan-17	Flagship	Neptune	15	13	2	-----	376
Ice Giant Study: Uranus Orbiter	JPL	Jan-17	Flagship	Uranus	15	12	3	-----	470
Ice Giant Study: Uranus Orbiter + Probe	JPL	Jan-17	Flagship	Uranus	15	12	3	-----	376
Triton Ocean World Surveyor	University of Arizona/APL	Jun-21	New Frontiers	Triton	21	16.7	4	538	325
Uranus Orbiter & Probe	NASA GSFC/UCSC/APL	Jun-21	Flagship	Uranus	18.5	13.4	5.1	735	-----

Appendix C.—Acronyms

APL	Applied Physics Laboratory
EODL	End of Design Life
ETG	Electrically heated Thermoelectric Generator
JPL	Jet Propulsion Laboratory
GPHS	General-Purpose Heat Source
GSFC	Goddard Space Flight Center
IV&V	Independent Verification and Validation
LES	Lincoln Experimental Satellite
M2020	Mars 2020
MHW	Multi-Hundred Watt
MMRTG	Multi-Mission Radioisotope Thermoelectric Generator
MSL	Mars Science Laboratory
RPS	Radioisotope Power Systems
RTG	Radioisotope Thermoelectric Generator
SNAP	Systems for Nuclear Auxiliary Power

